Governance-Aware Earth Observation Learning

Contract-Governed Training with Observed Service Agreement Graphs

Wenzhang Du
Bachelor of Engineering in Computer Engineering
Faculty of Engineering and Technology
Mahanakorn University of Technology
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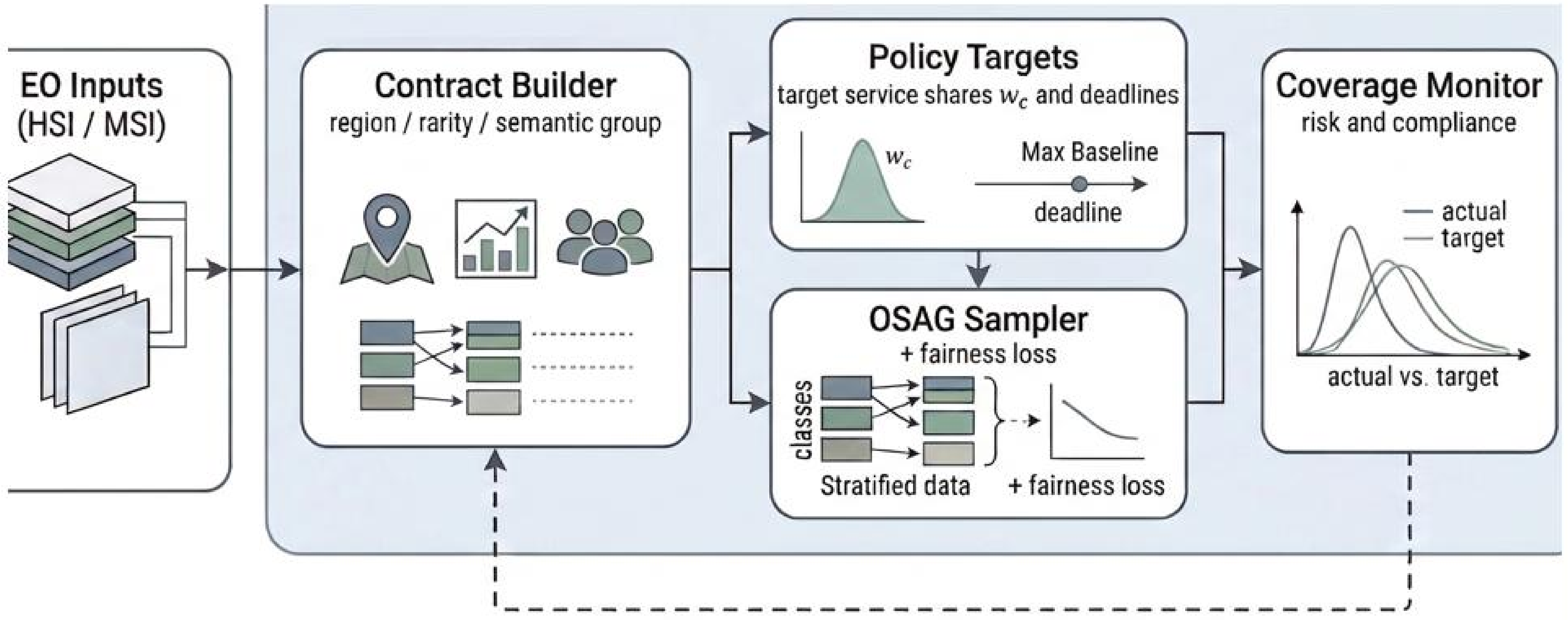
Governance-aware EO training with explicit contract-level service control.



Method overview visual

Talk Roadmap

This 15-minute defense moves from the problem setting to the method, then to fresh evidence, and finally to the dispatch demo and claim boundaries.

1. Problem and Motivation

Why accuracy-only EO training misses the service-allocation question.

2. OSAG Framework and Benchmarks

Contracts, target shares, PrioCovErr, and the HSI/MSI evaluation setup.

3. Evidence: Results, Ablation, and Robustness

Fresh five-seed results, trade-offs, contract design, runtime, and scoped backbone checks.

4. Dispatch Demo, Limitations, and Reproducibility

Operational behavior, claim boundaries, and the appendix-backed evidence chain.

The roadmap is short on purpose: it previews the flow without adding a separate personal-introduction slide.

Background and Motivation

Why the standard training view is incomplete

- Standard EO training usually optimizes aggregate predictive utility such as overall accuracy.
- Real EO services care about who receives attention during training: rare crops, vulnerable regions, or policy-critical semantic groups.
- If service allocation stays implicit, a model can look strong globally while underserving the contracts that matter most.
- This thesis treats training exposure as a governance object rather than as a side effect of data frequency.

Problem in one sentence

The key question is not only 'Does the model perform well on average?' but also 'Who is being served during training?'

Why governance is needed

Accuracy-only view

good global score

Can still ignore policy-critical strata if they are statistically small.

Governance view

explicit service

Tracks whether observed exposure matches the intended contract policy.

This thesis adds a lightweight governance layer instead of replacing the classifier backbone.

Research Gap and Guiding Questions

Central gap

EO training pipelines rarely expose service allocation explicitly. They optimize losses over samples, not compliance with a policy over contracts.

RQ1

How can EO training be reformulated so that service allocation over meaningful contracts becomes explicit and measurable?

RQ2

Can a lightweight governance layer drive empirical coverage toward target shares without requiring a new backbone?

RQ3

What is the empirical relationship between governance quality and predictive utility on HSI and MSI benchmarks?

RQ4

Can the thesis provide a runnable demo that communicates the governance mechanism clearly in a defense setting?

The deck follows the same arc: problem, method, fresh evidence, operational demo, and clear claim boundaries.

Thesis Contributions

Conceptual

Reframes EO training as a service-allocation and governance problem rather than an accuracy-only optimization problem.

Methodological

Develops OSAG as a contract-aware governance layer with target service shares, empirical coverage tracking, alpha mixing, and lambda_C fairness regularization.

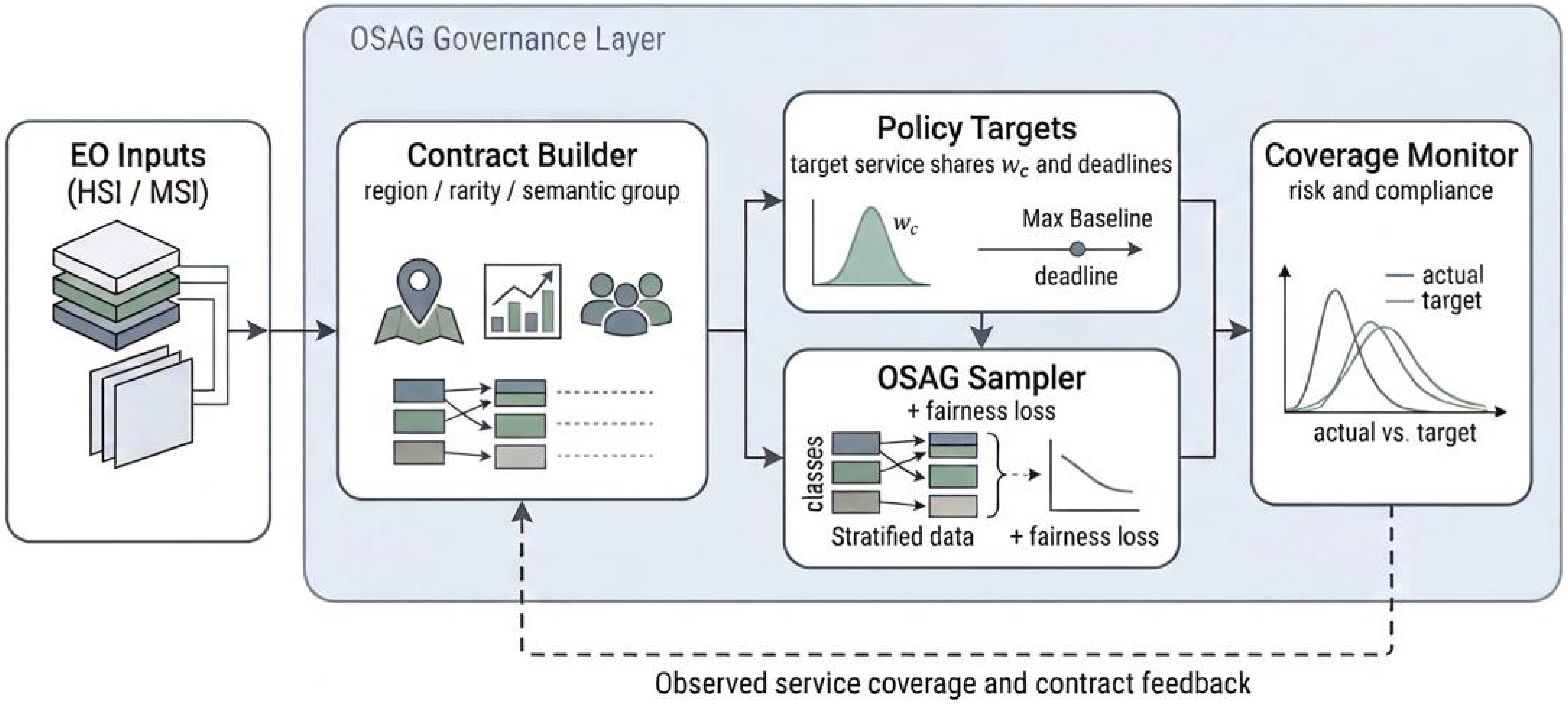
Empirical

Provides fresh five-seed reruns on corrected Indian Pines + Salinas and EuroSAT MSI, plus contract ablation, runtime, and a scoped ResMLP robustness extension.

Artifact-level

Adds a dispatch demo and a reproducibility appendix so the thesis can be defended through both benchmark evidence and a runnable operational artifact.

Figure 1.1: Method Overview



How to read the pipeline

- EO inputs provide sample metadata, labels, and contract-relevant attributes.
- The Contract Builder groups samples into service units such as grid cells, semantic groups, or rarity-aware groups.
- Policy Targets convert priorities and contract sizes into explicit target service shares.
- The OSAG Sampler injects the policy into training exposure, while the Coverage Monitor measures compliance.
- The feedback loop makes service allocation visible, auditable, and controllable during ordinary training.

Key point: OSAG wraps a standard EO classifier. It does not replace the backbone.

Core Concepts

Contract

A meaningful service unit, not just a class label. Example: dataset + grid cell + rarity flag.

Target service share

The intended fraction of training attention assigned to each contract.

Empirical coverage

The observed fraction of training steps actually spent on each contract.

PrioCovErr

The L1 distance between empirical coverage and target shares. Smaller means better policy compliance.

alpha

Mixing knob between the baseline sampler and pure OSAG. Higher alpha means stronger governance control.

lambda_C

Fairness-loss weight that adds pressure when high-priority contracts remain harder in loss space.

Benchmarks and Contract Construction

Benchmarks

- HSI main benchmark: corrected Indian Pines + corrected Salinas, aligned to a common spectral subset and merged into a joint hyperspectral setting.
- MSI main benchmark: canonical 13-band EuroSAT MSI extracted from dataset/EuroSAT_MS.zip.
- Main tables are fresh five-seed reruns. The EuroSAT contract-design ablation is a fresh three-seed rerun.

Contract schemas

Case	Schema	Count
HSI	dataset + 4x4 grid cell + rare-class flag	44
EuroSAT coarse	four broad semantic groups	4
EuroSAT fine	semantic group + training-split rare flag	6

Target-share rule

Target share is set as w_c proportional to p_c * n_c, where p_c is contract priority and n_c is contract size.

- HSI: rare contracts use priority 3; others use priority 1.
- EuroSAT coarse: urban and water use priority 2; agri and natural use priority 1.
- EuroSAT fine: rare contracts use priority 2; non-rare contracts use priority 1.

Main Fresh Five-Seed Benchmark Results

Defense-friendly summary of the central operating points. All numbers are means; values are percentages except PrioCovErr.

HSI: Indian Pines + Salinas

Policy	Acc_all	Acc_high	PrioCovErr
Random	91.47	82.97	23.49
Class-Balanced	90.25	86.26	31.37
OSAG-Mix (a=0.50)	91.68	86.51	11.75
OSAG	90.85	88.83	0.26
OSAG-FairLoss (l=0.5)	91.10	91.15	0.26

MSI: EuroSAT

Policy	Acc_all	Acc_high	PrioCovErr
Random	86.93	73.68	23.81
Class-Balanced	85.91	74.25	17.18
OSAG-Mix (a=0.75)	86.70	74.75	5.96
OSAG	86.39	76.05	0.20
OSAG-FairLoss (l=0.5)	86.30	77.04	0.20

Main benchmark message

PrioCovErr collapses toward zero

Pure OSAG and OSAG-FairLoss move policy misalignment from roughly 24 points to about 0.2-0.3 points on both benchmarks.

Utility remains competitive

Acc_all stays close

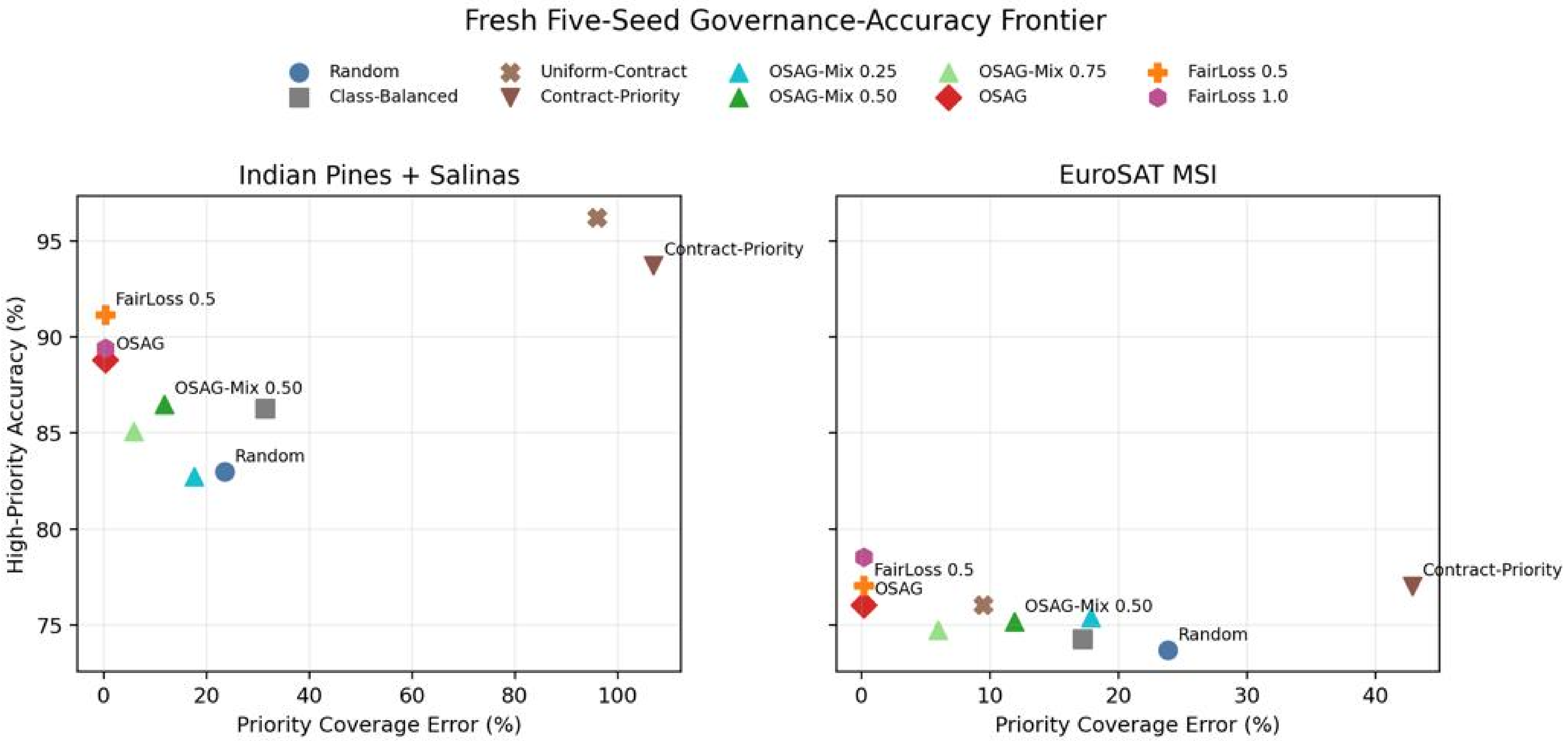
The governance gain is achieved without abandoning useful predictive performance.

High-priority utility

Acc_high improves

OSAG-family operating points improve or preserve high-priority performance while aligning service exposure.

Why Accuracy Alone Is Not Enough



Read Table 5.2 correctly

The reported deltas are absolute percentage-point differences relative to Random, not relative percentage changes.

Two benchmark examples

- HSI OSAG vs Random: Acc_all -0.62, Acc_high +5.85, PrioCovErr -23.23 points.
- EuroSAT OSAG vs Random: Acc_all -0.54, Acc_high +2.36, PrioCovErr -23.61 points.
- Heuristics can look good on Acc_high alone, yet still overserve some contracts and violate the target policy badly.

Takeaway

The thesis contribution is governance-aware training, not raw Acc_all domination.

Contract Design, Runtime, and Robustness

EuroSAT contract-design ablation

Fresh three-seed rerun. Fine contracts preserve more accuracy while still reaching near-zero PrioCovErr.

Variant	Rnd Acc_all	OSAG Acc_all	Rnd PrioCovErr	OSAG PrioCovErr
Coarse	86.98	86.03	33.29	0.12
Fine	86.98	86.28	23.80	0.21

- Fine contracts start from a lower Random PrioCovErr baseline.
- The accuracy cost of governance is smaller in the fine design than in the coarse design.
- This supports the thesis claim that contract design modulates governance cost.

Runtime interpretation

EuroSAT per epoch

Random 1.367 s | OSAG 1.358 s

The practical overhead is comparable.

HSI per epoch

Random 3.524 s | OSAG 2.938 s

Random is slower in this rerun mainly because of one outlier seed, not because OSAG is intrinsically faster.

Fairness penalty

modestly higher cost

OSAG-FairLoss stays in the same practical range but is somewhat more expensive than pure OSAG.

Thesis wording: runtime should be interpreted as comparable-overhead evidence, not as a speed claim.

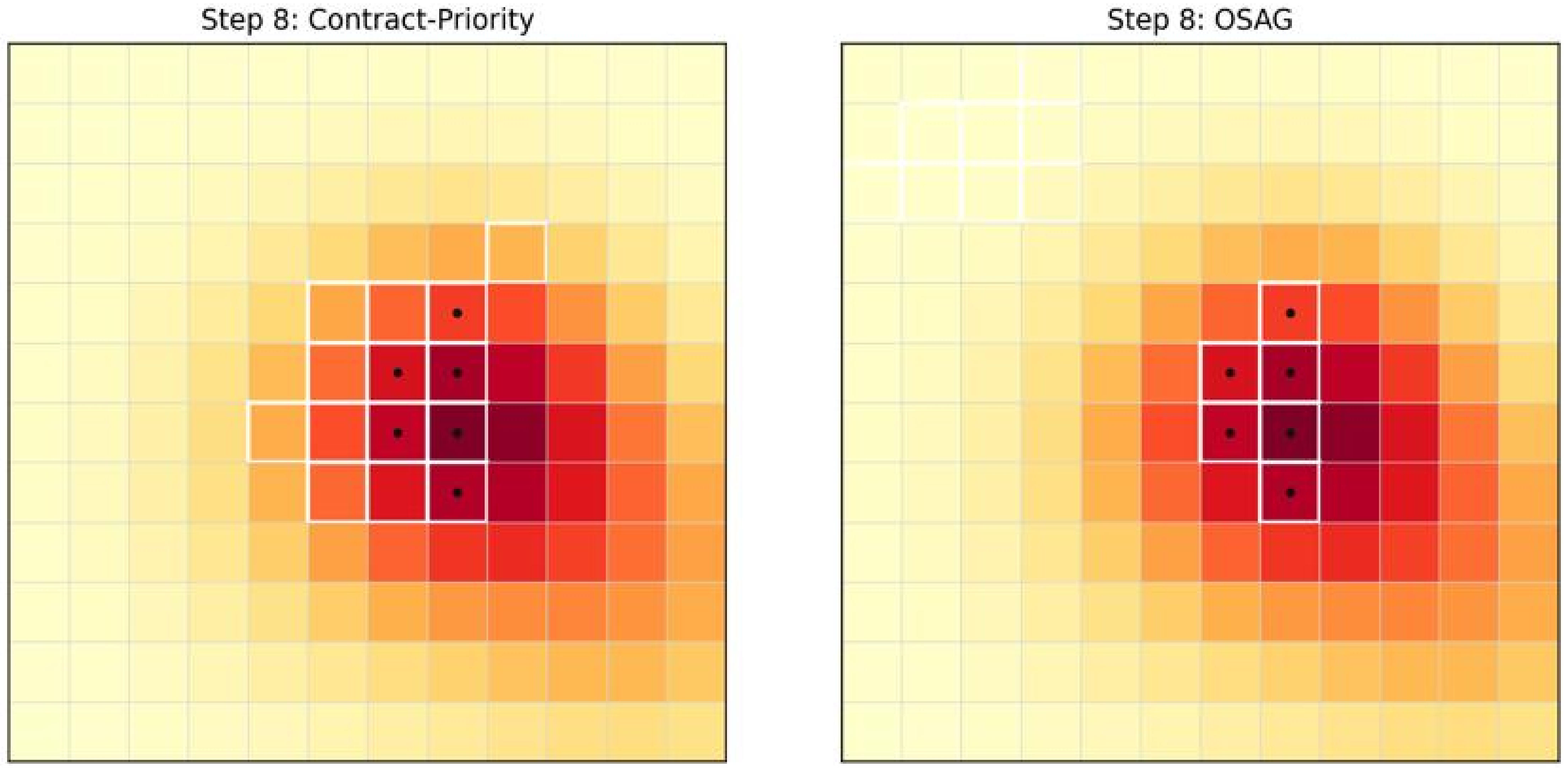
Scoped backbone robustness

EuroSAT only. ResMLP is a robustness extension, not a replacement for the main benchmark design.

Policy	Backbone	Acc_all	Acc_high	PrioCovErr
Random	Simple	86.93	73.68	23.81
Random	ResMLP	86.34	73.51	23.79
OSAG	Simple	86.39	76.05	0.20
OSAG	ResMLP	86.26	74.10	0.15

- Absolute utility shifts slightly with the stronger backbone.
- The core governance conclusion survives: OSAG still keeps PrioCovErr near zero.
- The thesis is not claiming state-of-the-art backbone performance.

Why a Dispatch Demo Is Needed



Operational scenario

- 12 x 12 emergency grid, 10 decision steps, 6 service contracts.
- Observation budget = 14 tiles per step; annotation budget = 6 tiles per step.
- High-priority contracts face short deadlines and revisit pressure.
- Compared policies: Random, Contract-Priority, and OSAG.

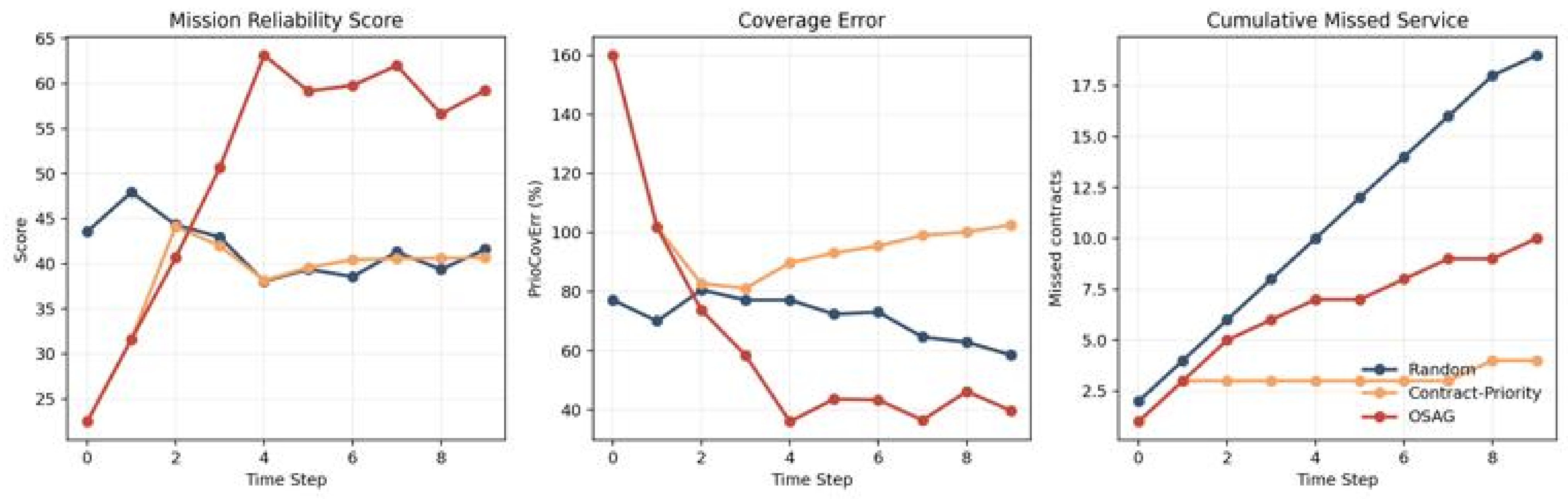
Why it belongs in the thesis

- The real benchmark is the scientific evidence chain.
- The demo makes the governance behavior visible under budgets, deadlines, and missed-service consequences.
- It supplements the benchmark; it does not replace it.

Dispatch Demo Results

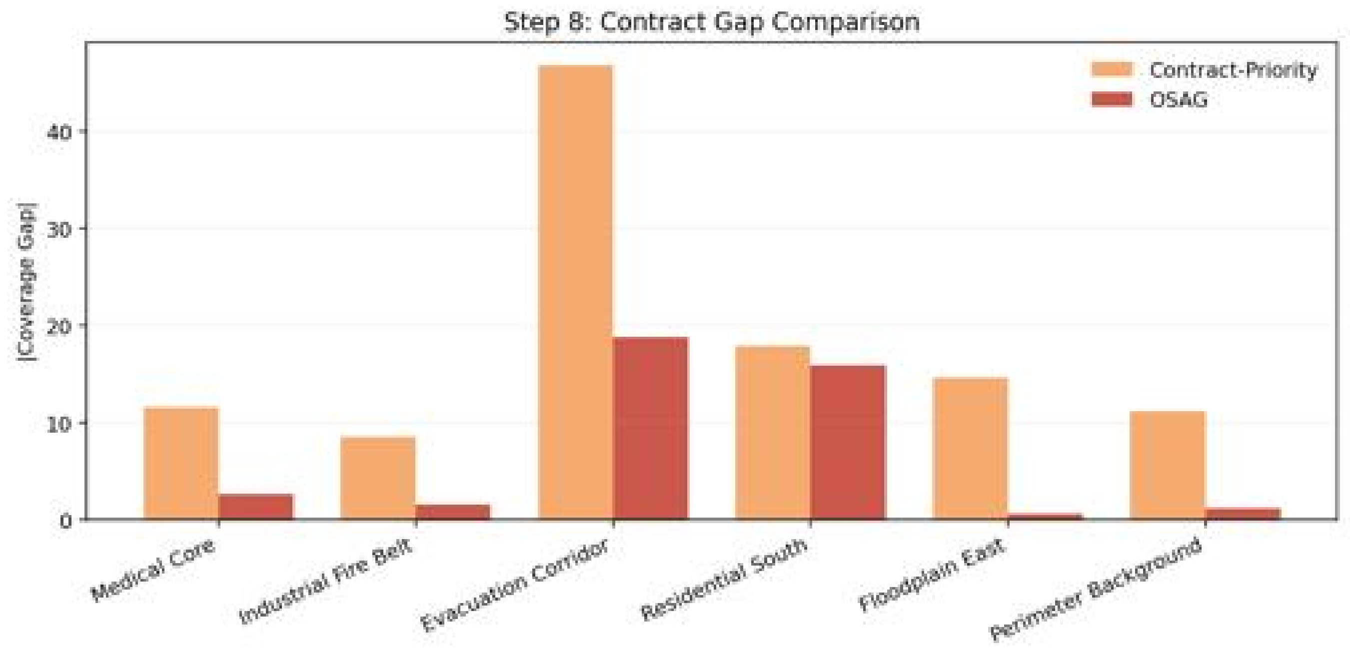
End-of-run summary

Policy	Reliability	Compliance	Coverage Err	Missed	Q_high
Random	41.60	53.67%	0.586	19	82.68
Contract-Priority	40.68	22.91%	1.026	4	80.43
OSAG	59.23	60.96%	0.397	10	81.47



How to interpret the demo

- OSAG is strongest on the governance-oriented summary: reliability, compliance, and coverage error.
- Contract-Priority looks attractive only if the user over-focuses on missed service or critical capture alone.
- The saved sensitivity audit shows OSAG remains top-ranked under equal weights and several local weight perturbations.
- Only more extreme capture-dominated weightings favor Contract-Priority.



Limitations and Claim Boundaries

What the thesis supports

- Governance-aware sampling can strongly reduce contract-level policy misalignment on the current HSI and EuroSAT MSI benchmarks.
- The effect survives fresh reruns, a contract-design ablation, runtime scrutiny, a scoped ResMLP extension, and an operational demo.
- The reproducibility appendix makes the evidence chain auditable through scripts, configs, manifests, logs, and saved outputs.

What is not being claimed

- The HSI split is pixel-stratified rather than spatial-blocked, so absolute accuracy may be optimistic relative to strict deployment conditions.
- The 'graph' is a modeling view, not a full explicit graph optimizer in the current implementation.
- The main benchmark uses a lightweight MLP to isolate governance effects; the thesis is not a backbone SOTA study.
- The ResMLP extension is scoped to EuroSAT only.
- The demo is supportive evidence, not the main benchmark.
- Provenance is documented through manifests and stage reports rather than a git commit ledger inside the thesis workspace.

Reproducibility and Public Release

Fully rerun main benchmark

HSI and EuroSAT MSI main tables are fresh five-seed reruns from formal scripts and canonical local inputs.

Fully rerun extensions

EuroSAT coarse-vs-fine ablation is a fresh three-seed rerun; the EuroSAT ResMLP extension is a fresh five-seed rerun.

Artifact-level reproducibility

Dispatch demo outputs, thesis figures/tables, and the thesis PDF can all be regenerated locally.

Evidence boundary and companion release

- Historical conference CSV files remain in the workspace only for traceability checks. They are not presented as fresh evidence.
- The thesis workspace is not a git repository, so provenance is documented through manifests, configuration snapshots, saved outputs, and stage reports.
- Companion public source release: https://github.com/dqswordman/osag-eo-governance

Final Takeaways

1. The thesis contribution is governance-aware EO training.

The key idea is to make service allocation explicit through contracts and target shares.

2. OSAG reduces policy misalignment sharply.

Fresh five-seed reruns show that PrioCovErr drops from roughly 24 points to about 0.2-0.3 points on both benchmarks.

3. The evidence chain is broader than a single table.

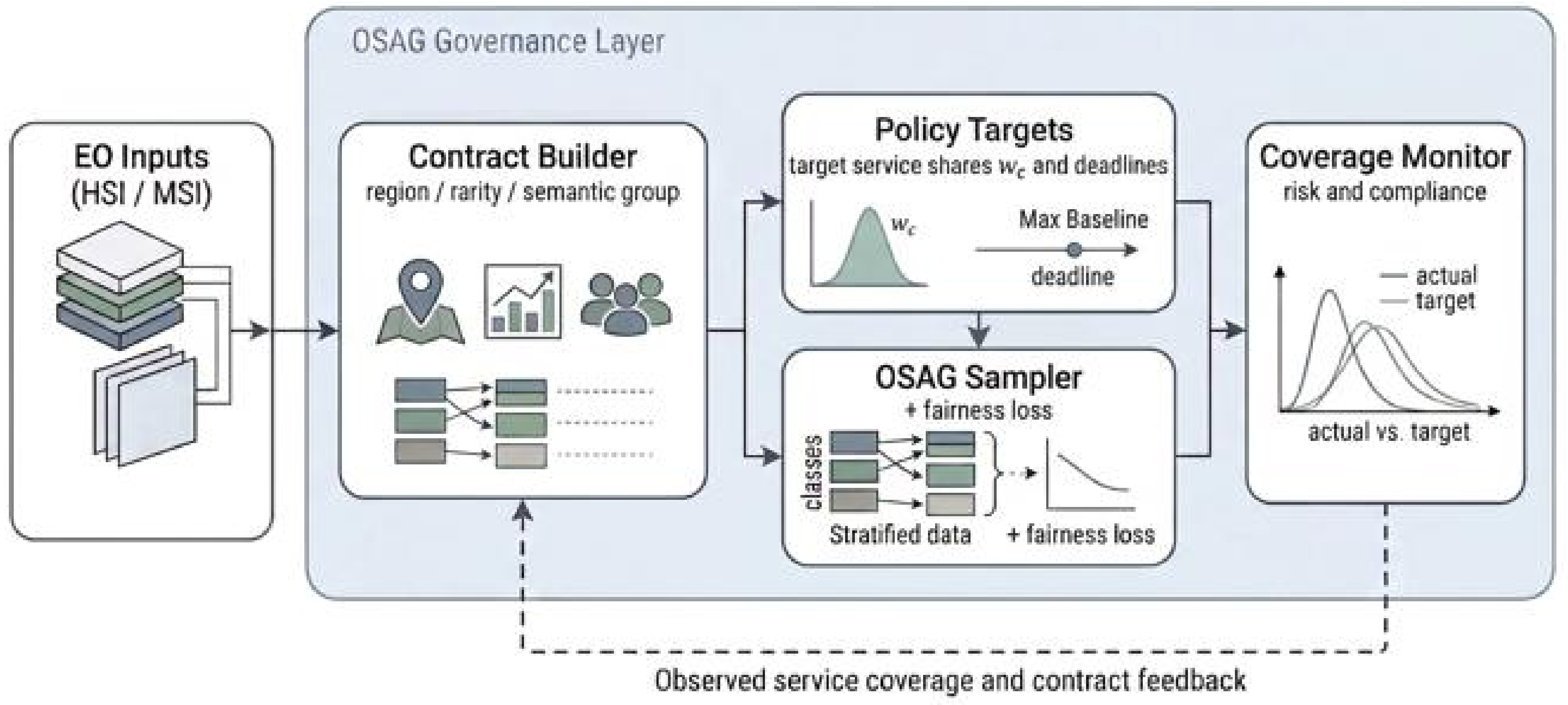
Benchmark reruns, contract ablation, runtime/robustness checks, dispatch demo, and Appendix A together form the defense story.

Natural next step: evaluate the same governance layer under stricter spatial-blocked splits and richer contract graphs.

Governance-Aware Earth Observation Learning

Observed Service Agreement Graphs offer a lightweight way to make EO training policy-aware, auditable, and defense-ready.

Questions and comments are welcome.



Companion public release: github.com/dqswordman/osag-eo-governance

Backup: Where is the 'graph' really?

- In the current thesis, the graph is a modeling view over the contract space, not a separate graph neural network or explicit graph optimizer.
- The graph intuition matters because contracts can be related through semantic similarity, spatial adjacency, or hierarchical refinement.
- Chapter 3 uses this graph view to motivate why contract design changes governance cost, especially in the EuroSAT coarse-vs-fine study.
- Future work could instantiate the graph explicitly through region adjacency graphs or learned semantic neighborhoods.

Safe oral answer

graph = structured contract space

The current implementation is mainly a contract-aware sampling and monitoring layer.

Do not overclaim

not a full graph optimizer

The thesis deliberately treats explicit graph algorithms as future work.

Backup: HSI Split and Spatial Leakage Boundary

What the pipeline does

- Corrected Indian Pines and corrected Salinas are aligned to a common spectral subset and merged into a joint HSI setting.
- Labeled pixels are flattened first, then assigned by a class-stratified 60/20/20 train/validation/test split.
- This is pixel-stratified, not spatial-blocked.

How to answer the leakage question

- Yes, neighboring pixels from the same original scene can appear in different splits.
- That means absolute accuracy may be optimistic relative to a stricter spatial-blocked evaluation.
- The main thesis claim should therefore be read as a comparative governance result under a fixed benchmark protocol.
- The benchmark still remains useful because all compared policies share the same split and the governance effect is large and consistent.

Backup: Reproducibility Scope

Level	What is included	Current boundary
Fully rerun main benchmark	HSI + EuroSAT MSI main tables over 5 seeds	scripted reruns from canonical local inputs
Fully rerun extensions	EuroSAT contract ablation over 3 seeds; ResMLP extension over 5 seeds	scoped extensions, not a full backbone study
Artifact-level reproducibility	demo outputs, thesis-ready figures/tables, thesis PDF	can be regenerated locally

Important provenance note

- Historical conference CSV files are retained only for traceability comparisons.
- The thesis workspace itself is not a git repository.
- Provenance is therefore documented through manifests, configs, logs, stage reports, and saved outputs rather than commit hashes.

Backup: Why Lightweight MLP and Only ResMLP Extension?

Why a lightweight MLP?

- The thesis studies a governance layer, so the main benchmark intentionally keeps the backbone simple and interpretable.
- A lightweight MLP makes it easier to attribute changes to the governance mechanism instead of to backbone complexity.
- This choice avoids overstating the work as a state-of-the-art architecture study.

Why only a ResMLP extension?

- The EuroSAT ResMLP experiment is a scoped robustness check, not a second main benchmark.
- It asks whether the governance conclusion survives a modest backbone strengthening.
- The answer is yes: OSAG still keeps PrioCovErr near zero, even though the best policy-backbone pair depends on the metric.

Backup: Demo Metrics, Q_high, K, and Sensitivity

Metric interpretation

- Reliability is a hand-set composite of compliance, coverage error, missed service, and critical capture K.
- Q_high is separate: it is the final mean high-priority model-quality state, not the same quantity as K.
- Coverage error in the demo is an operational contract-level metric analogous to, but not identical with, benchmark PrioCovErr.
- Reliability is useful as a dashboard summary, but it does not replace the primitive metrics.

Sensitivity audit

- No new simulation was rerun for the sensitivity note.
- The audit recomputed rankings from saved dispatch CSV outputs only.
- OSAG stays top-ranked under equal weights and several local perturbations around the thesis weighting.
- Only more extreme capture-dominated weightings favor Contract-Priority.

Backup: Companion Public Source Release

Repository

https://github.com/dqswordman/osag-eo-governance

- Packages the cleaned rerun scripts, configuration template, dispatch demo, and representative result snapshots.
- Matches the thesis narrative: governance-aware EO training, fresh rerun pipeline, and operational dispatch demo.
- Useful in defense as a companion release, but the thesis itself remains grounded in the locked local artifact chain.

Repository area	Purpose
scripts / configs	formal rerun pipeline and configuration template
osag_demo	dispatch simulation, asset generation, and dashboard
results snapshots	representative fresh outputs for documentation
README	project overview, setup, and usage instructions